Dimensionality Reduction

- Principal Component Analysis (PCA)

```
from sklearn.decomposition import PCA
pca = PCA(n_components = n)
x_pca = pca.fit_transform(x)
explained_variance = pca.explained_variance_ratio_
```

- Linear Discriminated Analysis (LDA)

```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis as LDA
lda = LDA(n_components = n)
x_lda = lda.fit_transform(x,y)
```

- Kernel PCA

```
from sklearn.decomposition import KernelPCA
kpca = KernelPCA(n_components = 2, kernel = 'rbf',gamma=0.1)
x_kpca = kpca.fit_transform(x)
```

- PCA vs Kernel PCA vs LDA

Feature	**PCA**	**Kernel PCA**	**LDA**
Full Name	Principal Component Analysis	Kernel PCA	Linear Discriminant Analysis
Type	Unsupervised	Unsupervised	Supervised
Uses Target (y)?	❌ No	❌ No	✅ Yes
Linearity	Linear	Non-linear	Linear
Goal	Maximize variance	Capture non-linear structure	Maximize class separation
Best For	General reduction	Complex/non-linear data	Classification problems

Feature	PCA	Kernel PCA	LDA
Output Limit	≤ features	≤ samples	≤ classes − 1
Kernel Used?	❌ No	✅ Yes	❌ (basic LDA)

PCA dimensionality reduction: n_components is 2,5,10

```
result
```

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
PCA	0.75	0.694444	0.833333	0.805556	0.833333	0.694444	0.75

```
result
#5
```

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
PCA	0.972222	0.972222	0.972222	0.944444	1.0	0.888889	0.944444

```
result
#10
```

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
PCA	1.0	1.0	1.0	0.944444	1.0	0.888889	0.861111

LDA dimensionality reduction:

n_components<=(number of classes-1) so here number of classes is 2 so n-components not exceed more than 2.

It is supervised learning need input and output variables

```
result
#2
```

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
PCA	1.0	1.0	1.0	0.972222	0.916667	0.916667	0.972222

Kernel_PCA:

```
result
#2
```

	Logistic	SVMl	Decision	Random
PCA	0.6375	0.65	0.55	0.55

```
result
#5
```

	Logistic	SVMl	Decision	Random
PCA	0.6375	0.65	0.6625	0.6375

```
result
#10
```

	Logistic	SVMl	Decision	Random
PCA	0.6125	0.65	0.55	0.6375